

CHE 311 Homework Week 12

1. For flow in a circular tube, please evaluate the ratio of $\langle V_z^2 \rangle / \langle V_z \rangle^2$ for (1) laminar flow (using parabolic velocity profile) and (2) turbulent flow (using 1/7-power expression).
2. For turbulent flow in smooth circular tubes, the above 1/7-power is not general. Empirical fitting of the curves of the time-smoothed velocity profiles suggested the following function,

$$\frac{\bar{v}_z}{\bar{v}_{z,max}} = \left(1 - \frac{r}{R}\right)^{1/n},$$

in which n depends on Reynolds number: near $Re = 4000$, $n=6$; near $Re = 1.1 \times 10^5$, $n=7$; and near $Re = 3.2 \times 10^6$, $n=10$. Show that the ratio of average to maximum velocity is,

$$\frac{\langle \bar{V}_z \rangle}{\bar{v}_{z,max}} = \frac{2n^2}{(n+1)(2n+1)}.$$